Combinatorics 8

Siddharth K

November 2025

Problem 1

Part (a)

Let (a_i) be such a sequence. Then for each a, take a right step and for each b, take an up step. Since the number of a's up to some point will always be at least the number of b's, the number of right steps will be at least the number of up steps, so the path stays under the line $y = x$.

Now, take a Dyck path, which can be notated as a sequence of Rs and Us. Let $a_i = a$ if the corresponding step in the Dyck path is R and $a_i = b$ otherwise. Then the number of a's will always be greater than the number of b's up to some point.

Problem 2

Let T_n be the number of binary trees with $n+1$ leaves. We can form such a tree by taking a left subtree with $k+1$ leaves and a right subtree with the remaining $n-k$ leaves. The number of such left binary trees is T_k and right binary trees is T_{n-k-1}. Since $k+1$ can range from 1 to n, the total sum is

$$T_n = \sum_{k=0}^{n-1} T_k T_{n-k-1},$$

which is the Catalan recurrence.

Problem 3

$$\sum_{m,n\geq 0} \binom{m+n}{m} x^m y^n = \sum_{n\geq 0} y^n \sum_{m\geq 0} \binom{m+n}{n} x^m$$

$$= \sum_{n\geq 0} y^n (1-x)^{-n-1} = \frac{1}{1-x} \sum_{n\geq 0} \left(\frac{y}{1-x}\right)^n$$

$$= \frac{1}{1-x} \cdot \frac{1}{1-\frac{y}{1-x}} = \frac{1}{1-x} \cdot \frac{1-x}{1-x-y} = \frac{1}{1-x-y}.$$

Problem 6

Let a_n be the number of binary sequences of length n without any two adjacent 1's. If the first element is 0, then the remaining $n-1$ elements form a valid sequence of length $n-1$. If the first element is 1, the next must be 0, and the remaining $n-2$ elements is valid. So $a_n = a_{n-1} + a_{n-2}$. We also have that $a_1 = 2$ and $a_2 = 3$, so $a_n = F_{n+2}$.

Problem 7

Let a_n be the number of valid ternary sequences of length n, let x_n be the number of these sequences that end with 0, y_n end in 1, and z_n end in 2. From 0, the next term can be $0, 1$, or 2, but from 1, the next term can only be 0 or 2, and from 2, the next term can only be 0 or 1. So

$$x_n + 1 = a_n,$$
$$y_{n+1} = x_n + z_n,$$
$$z_{n+1} = x_n + y_n.$$

Also, $a_n = x_n + y_n + z_n$, so

$$\begin{aligned} a_{n+1} &= x_{n+1} + y_{n+1} + z_{n+1} \\ &= a_n + (x_n + a_n) \\ &= 2a_n + a_{n-1}. \end{aligned}$$

So for $n \geq 1$, $a_{n+1} = 2a_n + a_{n-1}$.

Now, let $A(x) = \sum_{n\geq 0} a_n x^n$. We have that

$$\sum_{n\geq 0} a_{n+1}x^n = 2\sum_{n\geq 0} a_n x^n + \sum_{n\geq 0} a_{n-1}x^n,$$
$$(1 - 2x - x^2)A(x) = a_0 + (a_1 - 2a_0)x = 1 + x,$$
$$A(x) = \frac{1+x}{1-2x-x^2}.$$

The roots of the denominator are $1 \pm \sqrt{2}$, so the closed form for a_n is of the form

$$a_n = A(1+\sqrt{2})^n + B(1-\sqrt{2})^n.$$

Plugging in $a_0 = 1$ and $a_1 = 3$ gives us

$$A = \frac{1+\sqrt{2}}{2}, \quad B = \frac{1-\sqrt{2}}{2},$$
$$a_n = \frac{1}{2}((1+\sqrt{2})^{n+1} + (1-\sqrt{2})^{n+1}).$$

Problem 14

Part (a)

$$A(0) = \sum_{n\geq 0} p_n 0^n = p_0 = P(X = 0).$$

$$A(1) = \sum_{n\geq 0} p_n = 1.$$

Part (b)

$$P(X \text{ even}) = \sum_{n\geq 0} p_{2n},$$

$$A(1) = \sum_{n\geq 0} p_n,$$

$$A(-1) = \sum_{n\geq 0} p_n(-1)^n = \sum_{n\geq 0} p_{2n} - \sum_{n\geq 0} p_{2n+1},$$

$$A(1) + A(-1) = 2\sum_{n\geq 0} p_{2n} = 2P(X \text{ even}),$$

$$P(X \text{ even}) = \frac{A(1) + A(-1)}{2}.$$

Part (c)

$$\mathbb{E}[X] = \sum_{n\geq 0} np_n = A'(1).$$

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2 - 2X\mathbb{E}[X] + \mathbb{E}[X]^2]$$

$$= \mathbb{E}[X^2] - 2(\mathbb{E}[X])^2 + (\mathbb{E}[X])^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2,$$

$$\mathbb{E}[X^2] = \mathbb{E}[X(X-1)] + \mathbb{E}[X] = A''(1) + A'(1),$$

$$\mathrm{Var}(X) = A''(1) + A'(1) - (A'(1))^2.$$